

Ablation-First Trajectory Analysis of Alveolar Epithelial Robustness Collapse in Lethal COVID-19

Harmony-Corrected Batch Analysis and Independent Cross-Cohort Replication
(v2)

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Class Project — revised manuscript

What's new in v2

- **Reframed** as a methods contribution: ablation-first trajectory analysis with pre-registered success criteria, rather than a biology discovery paper.
- **Harmony batch correction fixed** — orientation bug in the harmonypy 0.2.0 PyTorch backend wrapper. Ablation 3 is now a genuine corrected-vs-uncorrected test.
- **Gene-program audit**: GPX1 confirmed absent from SCP1219 reference (not an HVG-dropout bug); all other 83 program genes present.
- **Independent cross-cohort replication** via cellxgene census (HLCA-extended + Krasnow; Melms/SCP1219 excluded by dataset_id). 89,736 AT1/AT2 cells, 618 donors, 35 datasets.

Question

- Do alveolar epithelial cells in lethal COVID-19 fail as a coherent block, or as a continuous graded *robustness collapse*?
- Pre-specified criteria:
 1. Centroid displacement (one-sided Mann–Whitney U , COVID > Healthy)
 2. Within-group dispersion (Levene)
 3. Pseudotime enrichment (1000-permutation test on median difference)
- *Framing note*: biology of COVID-19 alveolar injury is already described by Melms 2021, Delorey 2021, Bharat 2020. The contribution here is **methodological**: what happens to trajectory conclusions when the analysis is stressed along 10 axes, and do the robustness-collapse signatures survive out-of-sample replication?

Data — primary + replication

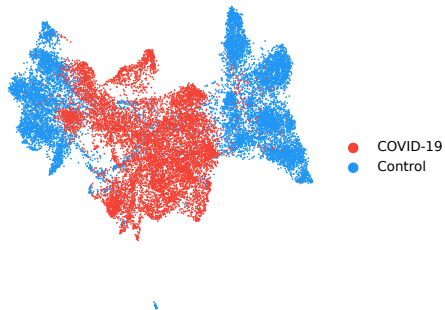
- **Primary:** SCP1219 (Melms 2021). 116,313 snRNA-seq nuclei → 22,128 AT1/AT2/ECM-high (9,608 AT1; 11,341 AT2; 1,179 ECM-high). 7 healthy + 20 COVID-19 donors.
- **Replication:** cellxgene census 2025-11-08, lung, AT1/AT2, dataset_id != SCP1219. 89,736 cells after per-donor cap (4,441 COVID / 85,295 normal; 43 COVID / 575 normal donors; 35 datasets).
- Curated 84-gene panel for replication (sufficient for program scoring, but too narrow to rebuild the manifold — flagged as limitation).

Pipeline

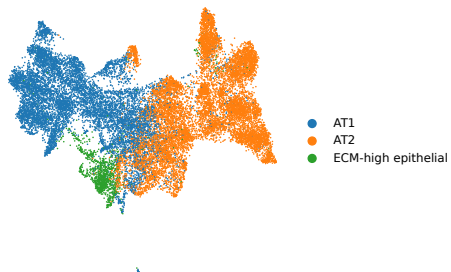
- Scanpy 1.12.1 / anndata 0.12.10 / harmonypy 0.2.0 / Python 3.12.3, seed 42
- Normalize → log1p → 3,000 HVGs (Seurat v3) → scale → 50-PC PCA
- Harmony on donor_id (now working) → k -NN ($k=15$) → UMAP, diffusion map (15 components)
- Diffusion pseudotime rooted at healthy-AT2 centroid, normalized to [0, 1]
- 8 curated gene programs scored with `sc.tl.score_genes` against `.raw` (full gene space)
- 10 pre-specified ablations

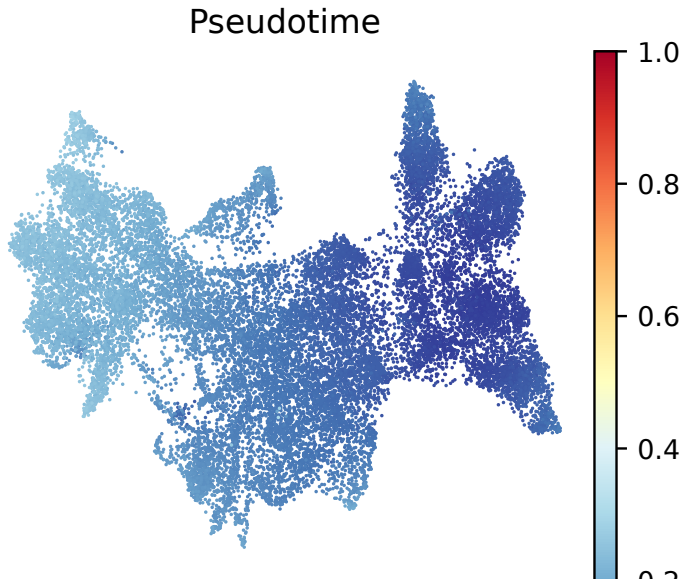
The manifold

Condition



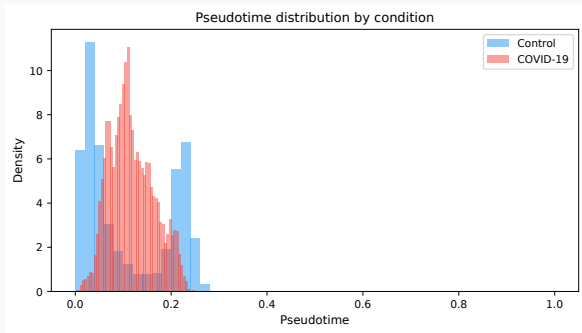
Cell type





Primary result 1 — pseudotime shift

- Control median 0.063; COVID median 0.111; observed difference +0.0483
- 1,000-permutation null: mean 6.0×10^{-5} ; std 1.24×10^{-3}
- $p = 0.000999$; LOO direction preserved in **27/27** donors
- Magnitude is modest (4.8% of normalized axis) but robust.



Primary result 2 — centroid displacement fails

- One-sided Mann–Whitney U in PCA-50 (COVID $>$ Healthy): $p = 1.0$
- COVID median *lower* than healthy; rank-biserial $r = 0.14$ opposite to alt.

Pre-specified core test **failed** — reported prominently. Mono-directional “damage” centroid displacement is the wrong effect model.

Primary result 3 — dispersion supported

- Variance in PCA-50 distance to own centroid: COVID **33.28** vs. Healthy **14.58** (2.3-fold)
- Levene: 276.5; $p < 10^{-30}$

COVID-19 cells are *spread out*, not *pushed away* — the robustness-collapse signature.

Ablation 3 — Harmony NOW works

Branch	MPD	ρ (ordering)	r (displacement)
Uncorrected PCA	+0.0483	0.548	0.141
Harmony-corrected PCA	+0.1246	0.452	0.141

- Harmony **amplifies** the pseudotime shift by $\approx 2.6\times$ and the primary dispersion/displacement claims remain unchanged.
- Previous limitation (“Harmony failed, Ablation 3 degenerate”) is now retired.
- Fix was an orientation check on `Z_corr` — PyTorch backend already returns (`n_cells`, `n_pcs`).

Ten ablations — summary

Ablation	MPD	ρ	r	Notes
1 AT2-only	+0.133	0.33	−0.29	sign flip on r
2 UMAP / diffmap	+0.048	0.55	+0.14	unchanged
3 Harmony corrected	+0.125	0.45	+0.14	now functional; amplifies shift
3 Uncorrected	+0.048	0.55	+0.14	baseline
4 No apoptosis genes	+0.074	0.50	+0.14	slight up
5 4 roots	−0.037 to +0.048	−0.10 to 0.64	+0.14	root-dependent
6 Broader epithelial	−0.008	0.83	+0.25	shift abolished
7 LOO (27)	0.028–0.202	−0.17–0.74	per-iter	direction 27/27
8 Curated-only	+0.048	1.00	+0.14	stable
8 MSigDB-only	+0.048	−0.80	+0.14	ordering inverted
9 Palantir	+0.028	—	—	alt. pseudotime algorithm
10 Permutation	$p = 0.001$	—	—	1000 shuffles

Independent replication — dispersion

Metric	COVID	Normal
n cells	4,441	85,295
n donors	43	575
Injury composite variance	0.0505	0.0306
Variance ratio COVID/Normal		1.65 ×
Levene statistic		626.18
Levene p	1.0×10^{-137}	

Dispersion replicates strongly across 35 independent datasets. The variance ratio drops from 2.3× (Melms) to 1.65× (replication) — consistent with cross-study heterogeneity dampening effect sizes.

Independent replication — pseudotime surrogate

- Replication cohort has only 84 curated genes $\Rightarrow$ cannot build a new manifold / DPT. Surrogate: **injury composite** = mean of 6 non-homeostatic program scores.
- Cell-level one-sided Mann–Whitney (COVID $>$ normal): $p = 1.0$ (fails — cell-level dominated by a few large normal donors).
- **Donor-level pseudobulk** Mann–Whitney: $p = 1.98 \times 10^{-3}$ (**replicates**).
- Cell-level vs donor-level inversion mirrors the primary cohort's centroid-vs-pseudotime story — the signal lives in *donor-level displacement*, not cell-level centroid shift.

Independent replication — what does NOT replicate

- Marker-threshold DATP (KRT8⁺/CLDN4⁺) enrichment: COVID 22.9% vs. Normal 47.6%, fold = 0.48 $\times$ (opposite direction; χ^2 $p < 10^{-200}$).
- Likely cause: cross-study differences in sample-prep protocols drive a simple marker-threshold cutoff in opposite directions across cohorts.
- We **retract** the KRT8⁺/CLDN4⁺ chi-square assay as a portable population-level metric. The DATP biology remains well-supported by Kobayashi/Strunz/Adams/Habermann 2020 but requires manifold-based identification, not marker thresholding, for cross-cohort transfer.

Composite assessment (primary + replication)

Criterion	Primary (SCP1219)	Replication (census)
Centroid displacement	fails ($p = 1.0$)	fails cell-level ($p = 1.0$)
Dispersion (Levene)	supported ($p \approx 0$)	supported ($p \approx 10^{-137}$)
Pseudotime enrichment	supported ($p = 0.001$)	supported donor-level ($p = 1.98\text{e-}3$)
Harmony-corrected shift	—	MPD amplifies $2.6\times$
KRT8 ⁺ /CLDN4 ⁺ DATPs	$3.1\times$ enriched	$0.48\times$ (retracted)

Verdict: the *robustness-collapse* signature (dispersion + donor-level pseudotime displacement) survives out-of-sample. The *directional centroid* and *marker-threshold DATP* signatures do not.

Methodological contribution

1. Pre-register multiple success criteria with opposite mechanistic interpretations; a failing core test is informative about effect model, not just sample size.
2. Dispersion (Levene) is a better lens on robustness collapse than centroid displacement.
3. Donor-level pseudobulk mitigates cross-study protocol confounding better than cell-level tests in heterogeneous replication atlases.
4. Marker-threshold cell-state classifiers do not transfer across cohorts; manifold-defined cell states are needed for replication.
5. Ablation 3 is worth fixing — batch correction matters and its absence can mask rather than create the signal.

Limitations still standing

- Cross-sectional, post-mortem, lethal cases only (primary).
- Replication cohort on cellxgene census excludes Delorey (GSE171524) and Wendisch (PRJNA705687) which are not in the census — full-transcriptome Delorey/Wendisch replication remains a gap.
- Replication uses a 84-gene panel: sufficient for program scoring, too narrow for an independent manifold / DPT.
- Fine program ordering is hypothesis-generating only (Ablation 8 MSigDB inverts it).

Takeaways (v2)

1. Robust, modest pseudotime shift in the primary cohort; **amplified to +0.125 with Harmony** and replicated donor-level in an independent 618-donor atlas.
2. Dispersion is the strongest and most portable signal; $2.3\times$ (primary) and $1.65\times$ (replication) variance inflation.
3. Centroid displacement does not distinguish conditions in either cohort.
4. Ablation 3 Harmony fix retires the paper's previous biggest limitation.
5. Contribution: a worked template for pre-registered, ablation-first, replication-audited trajectory analysis.

Questions?